

A L I V I N G A I D O S S I E R F O R Y O U R W A L L E T



Soulprint

Mint once. It evolves forever.

Every wallet has personality. Nobody can read it.

On-chain history holds an identity in raw counters — transactions, deploys, NFT mints, gas spent. But it's locked in numbers no human reads, and no contract composes on.

Meanwhile most “AI on-chain” projects today are smoke and mirrors: they call Gemini or Groq off-chain and only settle the result on-chain. The AI isn't really on-chain.

THE GAP

0

*on-chain primitives that compose on a
wallet's identity today*

(an MCP server, a gating contract,
an agent that asks “who is this?”)

An on-chain identity primitive, written by on-chain AI agents.

- 1 Point Soulprint at any wallet (yours or someone else's).
- 2 A Somnia JSON API agent reads its real on-chain activity.
- 3 A Somnia on-chain LLM (Qwen-3 30B, fixed seed, validator-consensus) writes a witty, structured dossier.
- 4 It mints as a soulbound NFT (ERC-5192). Can't be sold, can't be transferred.
- 5 An on-chain Cron keeps re-evaluating it — autonomously, forever.

End to end — about six seconds. AI executed by validators, output verified by consensus.

Try it. Now.

Everything below is live on Somnia Shannon testnet (chainId 50312).

FRONTEND

soulprint-psi.vercel.app

Mint a soulprint, look up any wallet.

GITHUB

github.com/DontFoldBB/soulprint

Public repo. 47 Hardhat tests green.

SOULPRINT CONTRACT

0x6876041cc67f9cd1b11e6e1827b13f3622d256e5

ERC-721 + ERC-5192 soulbound, dynamic on-chain tokenURI.

SOULPRINTCRON

0x0bf4e395ad3746632f86b5254fa18f0db3479d95

Somnia Reactivity handler. Self-rescheduling autonomy.

Architecture

A two-step on-chain agent pipeline with a cost-gate in the middle, an autonomy loop above.



The Soul Evolution System

On-chain visual evolution — derived from real activity, not from the LLM's mood.

10 stages × **3** archetype lines = **30** spirit forms

Stage 1 → 10

Bucketed from on-chain tx_count. Stage 4 unlocks at 75 tx, Stage 7 at 2 000, Stage 10 at 200 000.

Form lookup

(archetype × stage) maps to one of 30 distinct PNG spirit forms — Pathfinder Shade, Cartographer Spirit, DeFi Whale, Sybil Hydra, Onchain Legend...

Composable

stageOf / formIdOf / formSlugOf / evolutionOf are one-call reads. Stage + Form land in tokenURI traits — marketplaces and agents can index without parsing the dossier.

Prepaid Evolution Fuel

Solves the “autonomous tick eats the project's reserve forever” problem on-chain.

BEFORE

A shared reserve, drained forever

- One mint → autonomous evolution forever from the project's pocket.
- 100 active wallets ≈ 100 STT / day burn.
- Owner can withdraw the reserve — no protection for funded evolutions.

NOW (LIVE)

Per-soul fuel · anyone can keep a soul alive

- Mint earmarks INITIAL_FUEL_GRANT (0.4 STT ≈ 1 evolution) into the token's evoBalance.
- Each successful re-evolution deducts EVOLUTION_COST. Out of fuel → soul freezes (EvolutionPaused).
- ANYONE can call topUpEvolution(tokenId) — owner, fan, patron contract. STT locks per-token; owner cannot withdraw it (gated by availableForWithdraw, totalReserved is sacred).

Keeping a soul alive becomes a public good. Boost a friend. Boost a stranger.

Autonomy — proven live on-chain

Encode criterion #4 · Autonomous Performance

35

autonomous ticks

on the previous live cron

2 real evolutions · 33 cost-gated skips · 0 human tx

HOW

SoulprintCron is a Somnia Reactivity SomniaEventHandler.

- On each scheduled tick the handler calls `Soulprint.evolveBatch()` and then schedules the next tick — inside the same handler. The chain ticks the contract for us, on time, forever.
- Sender of every tick is the Reactivity precompile (address `0x...0100`). The contract owner's address has zero outgoing tx to the cron — fully verifiable on the explorer.
- Fresh cron at `0x0bf4...` is armed now; same code as the proven one — criterion holds.

Agent-first composability

Encode criterion #2 · the Soulprint is a primitive, not an end-product NFT.

ONE-CALL READS

```
profileOf(addr)
traitsOf(addr)
evolutionOf(addr)
evolutionFuel(tokenId)
```

Any contract or agent gets a wallet's Soulprint in one read — dossier, archetype, activity score, stage + form, fuel state.

MCP SERVER

```
node mcp/dist/index.js
get_soulprint(wallet)
```

Model Context Protocol server exposes the Soulprint to any AI agent over a standard protocol. Read-only, no key.

EXAMPLE GATE

```
ExampleGate.enter()
require(activityScore >= N)
```

Third contract gates access on Soulprint activity. Reputation that can't be bought, transferred, or faked.

How we map to the 4 judging criteria

1**FUNCTIONALITY**

Live on Somnia testnet. read → AI → soulbound mint pipeline runs end-to-end in ~6 s. 47 Hardhat tests green.

2**AGENT-FIRST DESIGN**

Orchestrates TWO Somnia base agents (JSON API + LLM) per pipeline. profileOf / traitsOf / evolutionOf / evolutionFuel for any contract to read. MCP server + ExampleGate as live consumers.

3**INNOVATION & TECHNICAL CREATIVITY**

Native on-chain inference (not off-chain). Chained read→reason agent pipeline. Soulbound IDENTITY (not collectible). On-chain dynamic tokenURI. Cost-gated evolution. 30-form Soul Evolution. Prepaid Evolution Fuel.

4**AUTONOMOUS PERFORMANCE**

SoulprintCron self-reschedules INSIDE the handler. ticks() rises, subscriptionId re-arms, zero human tx between ticks. Previous cron: 35 ticks → 2 real evolutions, all verifiable on explorer.

Tech stack & numbers

47

Hardhat tests

30

spirit PNG forms

10

evolution stages

~6s

end-to-end mint

0

human tx per evolution

THE STACK

Solidity 0.8.30

OpenZeppelin v5

ERC-721 + ERC-5192

viaIR · cancun EVM

Hardhat + viem

Somnia Agents

Somnia Reactivity

Qwen-3 30B (on-chain)

Model Context Protocol

Next.js 16 · Turbopack

Vercel

TypeScript 5.7



Soulprint

A composable identity primitive other agents read.

Cost-bounded by design. Soulbound. Self-evolving. Live on Somnia.

Mint once. It evolves forever.